



Karachi AQI Hub

AI-Powered Air Quality Forecasting & Health Intelligence

System Status

Champion Model: Gradient Boosting

Test R² Score: 0.9113

Test RMSE: 2.39 µg/m³

Hopworks Feature Store

Feature Group:

karachi_aqi_features (v1)

Selected Features: 18 Predictors

Sync Mode: Hourly GitHub Actions Cron

Refresh Live Data

Built for 10 Pearls Data Science Internship • Karachi, Pakistan

Karachi Air Quality Index & 72-Hour Forecaster

Karachi, Pakistan • Real-Time Satellite & Atmospheric Telemetry • Updated: Thursday, August 27, 2026 • 09:00 AM UTC

LIVE AQI STATUS: MODERATE



PM10 Coarse Dust

62.4 µg/m³

Temperature

29.1 °C

Relative Humidity

70.0 %

Wind Speed

19.4 km/h

Carbon Monoxide (CO)

193 µg/m³

Surface Pressure

1002 hPa

Health Advisory & Action Plan: Air quality is acceptable; moderate health concern for sensitive individuals.

72-Hour (3-Day) AQI Forecast | Exploratory Data Analysis & Trends | SHAP Explainability & Factor Insights | Scenario Simulator Sandbox | Model Leaderboard & MLOps Registry

Interactive Scenario Simulator (What-If Analysis)

Adjust meteorological and pollution parameters to simulate their impact on Karachi's forecasted air quality.

Current PM2.5 (µg/m³)

26.50

Temperature (°C)

29.10

Sulfur Dioxide SO2 (µg/m³)

4.60

PM10 Dust (µg/m³)

62.40

Relative Humidity (%)

70.00

Nitrogen Dioxide NO2 (µg/m³)

3.40

Carbon Monoxide CO (µg/m³)

193.00

Wind Speed (km/h)

19.40

Surface Pressure (hPa)

1002.50

Simulated Next-Hour PM2.5

26.5 µg/m³

MODERATE

Impact Interpretation:

Air quality is acceptable; moderate health concern for sensitive individuals.